

SMART INDIA HACKATHON 2026



TITLE PAGE

SANYOG संयोग — from problem, to pilot, to procurement

PROBLEM STATEMENT ID

SIH26136

PS CATEGORY

Software

THEME

Miscellaneous

TEAM ID

TEAM NAME

PRODHUB

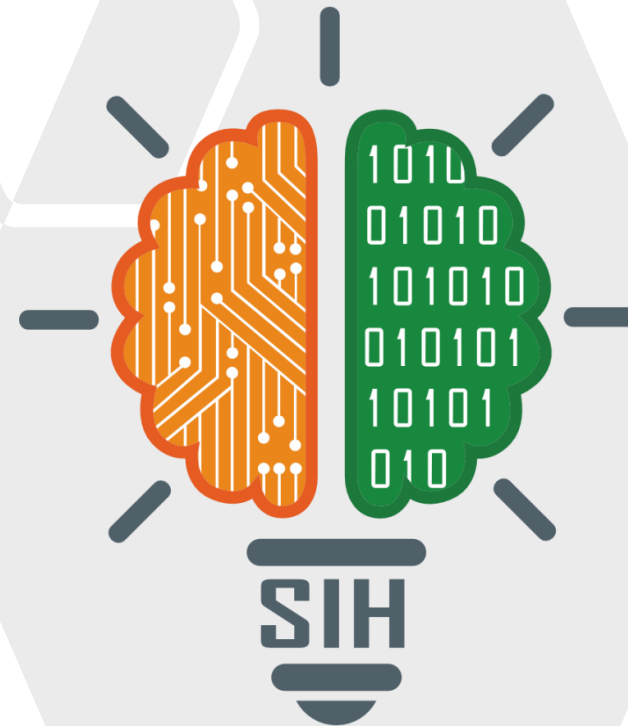
ORGANISATION

**Govt of Maharashtra ·
MSInS**

PROBLEM STATEMENT TITLE

Startup friendly public procurement mechanism that enables government departments to identify, pilot, procure, and scale innovative solutions from eligible startups

● **LIVE** sanyog.ashutosh-palai2005.workers.dev



IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

SANYOG (संयोग) — an end-to-end digital pathway that takes a government department from a written problem to a scaled, procured startup solution — in eight governed stages, on one platform.

1 · CHALLENGE

2 · SCREEN

3 · EVALUATE

4 · SANDBOX

5 · CONTRACT

6 · MILESTONES

7 · VALIDATE

8 · SCALE

Detailed explanation of the proposed solution

- Departments publish outcome-based challenges — a measured baseline → target is mandatory, so every pilot is verifiable by construction.
- DPIIT-recognised startups are auto-screened on application; pilots run in a governed sandbox with security clearance and a fixed exit date.
- Payments release 30/40/30 against verified milestones; independent validators record results against pre-locked KPIs.

How it addresses the problem

- Removes turnover, prior-experience and EMD barriers (GFR 173(i) & 170(i) as defaults) — quality criteria never waived.
- The three mandated documents — Problem Statement, Pilot MOU, Scale-Up Scorecard — are working digital forms, not PDFs.
- Demand becomes visible, payments timed, evaluations accountable, every decision evidence-based and audit-traceable.

Innovation and uniqueness of the solution

- Compliance is computed, not promised — the database refuses payment without verification, sandboxes without clearance, rejection without reason.
- KPIs are locked before measurement begins; deemed-approval clocks; attributable panel scores with mandatory written rationale.
- Three satisfactory departmental validations unlock scale-up — the evidence pack exports directly to GeM. No fresh tender.

Technologies to be used (e.g. programming languages, frameworks, hardware)

React 19 + TanStack Start

Cloudflare Workers · edge SSR

Supabase PostgreSQL · RLS

Groq LLM · FitScore™

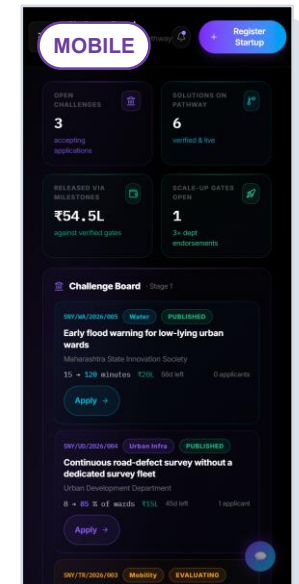
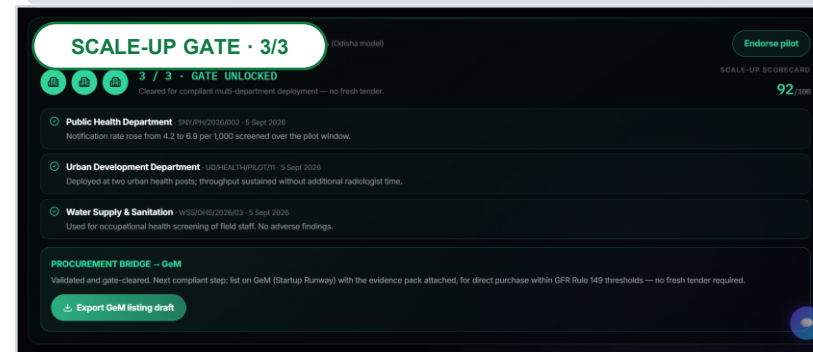
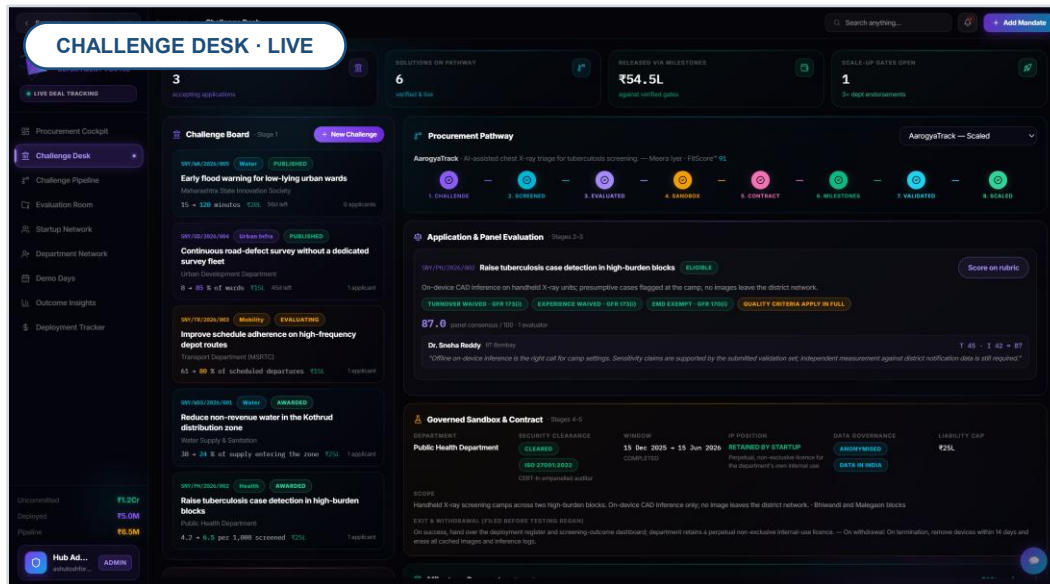
JWT + Google OAuth

DPIIT auto-screening

GeM · Startup Runway export

Rules enforced in the DB

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype — live at sanyog.ashutosh-palai2005.workers.dev · Challenge Desk · milestone engine · 3-department scale-up gate · mobile

Analysis of the feasibility of the idea

- **Needs no new law** — GFR 2017 already permits every relaxation applied — Rule 173(i) turnover & experience, Rule 170(i) EMD, Rule 172 capping the 30% advance; MSMED s.15/16 supplies the payment clock.
- **Policy demand exists** — Maharashtra Innovation Policy 2025, Section 13.7, commits to exactly these guidelines — Sanyog is the working mechanism, ready to adopt.
- **Precedents in force** — Telangana G.O. Ms. 08 (deemed approval) · Odisha's 3-client scale gate · Kerala GO(Ms) 2/2022 (₹50L direct route).
- **Prototype, not concept** — Fully built and LIVE — both portals, all 8 stages, the 3 mandated templates, screening, payments, validation and the scale gate.

Potential challenges and risks → Strategies for overcoming these challenges

⚠ Subjective, opaque evaluation

→ Dual-axis 0–50 scores stored per evaluator with mandatory written rationale and conflict declaration — defensible on audit.

⚠ Late payments starve startups

→ A 45-day clock on every tranche; overdue MSMED interest (3x bank rate) on the department's own screen; release blocked until verification.

⚠ Citizen-data & security exposure

→ CERT-In-empanelled audit before go-live; ISO/IEC 27001:2022 only; data anonymised and localised in India; fixed sandbox exit dates.

⚠ Pilots that never scale (iDEX: ~8%)

→ The 3-department gate converts pilot evidence into a GeM listing within GFR Rule 149 thresholds — no fresh tender.



IMPACT AND BENEFITS



Potential impact on the target audience

45 days

hard payment ceiling on every milestone (MSMED s.15)

16 weeks

from published challenge to a running pilot

3 validations

unlock multi-department scale — no fresh tender

100%

of decisions attributable and audit-traceable

STARTUPS

Paid in 45 days, not months · no turnover / experience / EMD walls · demand finally visible · one validated pilot opens every department and district.

DEPARTMENTS

Risk bounded by design (exit dates, liability caps, IP clarity) · decisions carry evidence · standard templates replace months of drafting.

CITIZENS

Water, health and mobility fixes reach the field in ~16 weeks · outcomes measured against locked baselines, not claimed in reports.

Benefits of the solution (social, economic, environmental, etc.)

ECONOMIC

Startup cashflow survives government sales — 30% mobilisation advance + clocked tranches; wider MSME participation in public procurement.

SOCIAL

Transparent, reasoned public decisions — arbitrariness engineered out; successful pilots scale to every district, not just the pilot ward.

ENVIRONMENTAL

Pilots target measurable resource loss — non-revenue water 38% → 24% in the demo arc; transformer failures pre-empted before outages.

GOVERNANCE

Every decision attributable and audit-traceable; scale-up rests on independently validated outcomes, not vendor claims.



RESEARCH AND REFERENCES



Details / Links of the reference and research work

General Financial Rules 2017 — Dept of Expenditure

Rules 149 (GeM thresholds), 170(i) EMD exemption, 172 (30% advance cap), 173(i) turnover & experience relaxation · [doe.gov.in](#)

MSMED Act 2006 — s.15, s.16

45-day payment ceiling that cannot be contracted around; default interest at 3x the RBI bank rate, compounded monthly

Maharashtra Innovation & Startup Policy 2025 — s.13.7

State commitment to startup-procurement guidelines and pilot support · [msins.in](#)

State precedents in force

Telangana G.O. Ms. No. 08 (deemed approval) · Odisha 3-client scale gate · Kerala GO(Ms) 2/2022/SPD (₹50L direct purchase)

CERT-In — Directions 2022 & Guidelines for Govt Entities 2023

Empanelled pre-hosting audit · 6-hour incident reporting · 180-day logs within India · [cert-in.org.in](#)

DPDP Act 2023 · ISO/IEC 27001:2022

Data-protection safeguards for sandbox pilots; the 2013 ISO edition expired 31 Oct 2025 — the platform accepts :2022 only

iDEX / DISC outcome data — Ministry of Defence

~548 challenges → ~45 procurement contracts (≈8%): the pilot-to-purchase gap Stage 8 is designed to close

US SBIR 15 U.S.C. 638(r)(4) · UK SBRI Condition 27

Competitive entry + verified pilot supports a lawful non-competed follow-on; arising IP retained by the supplier

NHS England KPI framework · HM Treasury Magenta Book

Baseline–target–threshold KPI schema and pre-registered evaluation design, adopted in Stages 1 and 7

Working prototype — Team PRODHUB · LIVE

[sanyog.ashutosh-palai2005.workers.dev](#) · full 8-stage platform, 3 mandated templates · code: [github.com/Ashutoshpalai3073/PRODHUB](#)